

Corrigé — Exercice 11

A. 1) $f'(x) = \frac{(x+1) - (x+2)}{(x+1)^2} = \frac{-1}{(x+1)^2} < 0$: f décroissante. **2)** $f(\sqrt{2}) = \frac{\sqrt{2}+2}{\sqrt{2}+1} = \sqrt{2}$; f décroissante donne $f([1, 2]) = [\frac{4}{3}, \frac{3}{2}] \subset [1, 2]$. **B. 3)** $u_0 = 1 \in [1, 2]$; si $1 \leq u_n \leq 2$, $u_{n+1} = f(u_n) \in [1, 2]$. **4)** $v_{n+1} = \frac{(1-\sqrt{2})(u_n-\sqrt{2})}{(1+\sqrt{2})(u_n+\sqrt{2})} = (2\sqrt{2}-3)v_n$. **5)** $|2\sqrt{2}-3| < 1$ donc $v_n \rightarrow 0$, d'où $u_n \rightarrow \sqrt{2}$.

